Symbiosis Institute of Technology, Pune

Faculty of Engineering
CSE- Academic Year 2026-27
Java Programming – Lab Batch 2024-28

Lab Assignment No:- 27	
Name of Student	Abishek Yadav
PRN No.	25070122014
Batch	A1
Class	CSE
Academic Year & Semester	2026-2027/3RD
Date of Submission	18/10/2026
Title of Assignment:	Develop a Java application demonstrating socket programming for communication between a client and a server.
Theory:	Socket programming enables communication between computers over a network using TCP/IP protocols. Java provides the Socket class for client-side communication and the ServerSocket class for server-side communication. A server listens for client requests on a specific port, while clients establish connections using the server's IP address and port number. Once connected, data is exchanged using input and output streams. Socket programming supports reliable communication between distributed systems and is widely used in chat applications, online gaming, file transfer systems, remote monitoring, and client-server architectures.
Practice Questions	1. Create a simple client-server chat application using sockets.

CODE:

```
import java.io.*;
import java.net.*;

public class ChatServer {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(5050);
            System.out.println("Server started.");
            System.out.println("Waiting for client...");
            Socket socket = serverSocket.accept();
            System.out.println("Client connected.");
            BufferedReader input = new BufferedReader(
                new InputStreamReader(socket.getInputStream())
            );
            PrintWriter output = new PrintWriter(
                socket.getOutputStream(), true
            );
            BufferedReader keyboard = new BufferedReader(
                new InputStreamReader(System.in)
            );
            String clientMessage;
            String serverMessage;
            while (true) {
                clientMessage = input.readLine();
                if (clientMessage == null ||
clientMessage.equalsIgnoreCase("bye")) {
                    break;
                }
                System.out.println("Client: " + clientMessage);
                System.out.print("Server: ");
                serverMessage = keyboard.readLine();
                output.println(serverMessage);
                if (serverMessage.equalsIgnoreCase("bye")) {
                    break;
                }
            }
            socket.close();
            serverSocket.close();
            System.out.println("Chat ended.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
```

```
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac ChatServer.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

C:\Users\Bina Computer\OneDrive\Documents\Java>javac ChatClient.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

C:\Users\Bina Computer\OneDrive\Documents\Java>java ChatServer
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

Server started.
Waiting for client...
Client connected.
Client: hello
Server: hello Client
Client: How are You ?
Server: I am fine
Chat ended.

C:\Users\Bina Computer\OneDrive\Documents\Java>
```

CODE:

```
import java.io.*;
import java.net.*;

public class ChatClient {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 5050);
            System.out.println("Connected to server.");
            BufferedReader input = new BufferedReader(
                new InputStreamReader(socket.getInputStream())
            );
            PrintWriter output = new PrintWriter(
                socket.getOutputStream(), true
            );
            BufferedReader keyboard = new BufferedReader(
                new InputStreamReader(System.in)
            );
            String clientMessage;
            String serverMessage;
            while (true) {
                System.out.print("Client: ");
                clientMessage = keyboard.readLine();
                output.println(clientMessage);
                if (clientMessage.equalsIgnoreCase("bye")) {
                    break;
                }
                serverMessage = input.readLine();
                System.out.println("Server: " + serverMessage);
                if (serverMessage.equalsIgnoreCase("bye")) {
                    break;
                }
            }
            socket.close();
            System.out.println("Chat ended.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>java ChatClient
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

Connected to server.
Client: hello
Server: hello Client
Client: How are You ?
Server: I am fine
Client: bye
Chat ended.

C:\Users\Bina Computer\OneDrive\Documents\Java>
```

QUESTION

Create a Java program using sockets where the server receives a message from the client and sends the same message back (Echo Server and Echo Client).

CODE:

```
import java.io.*;
import java.net.*;

public class EchoServer {
```

```
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(6000);
            System.out.println("Echo Server started.");
            System.out.println("Waiting for client...");
            Socket socket = serverSocket.accept();
            System.out.println("Client connected.");
            BufferedReader input = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            PrintWriter output = new PrintWriter(
                socket.getOutputStream(), true);
            String message;
            while ((message = input.readLine()) != null) {
                System.out.println("Client: " + message);
                output.println(message);
                if (message.equalsIgnoreCase("bye")) {
                    break;
                }
            }
            socket.close();
            serverSocket.close();
            System.out.println("Echo server stopped.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac EchoServer.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

C:\Users\Bina Computer\OneDrive\Documents\Java>javac EchoClient.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

C:\Users\Bina Computer\OneDrive\Documents\Java>java EchoServer
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

Echo Server started.
Waiting for client...
Client connected.
Client: Hello Server
Client: Java Socket Programming
Client: bye
Echo server stopped.

C:\Users\Bina Computer\OneDrive\Documents\Java>
```

CODE:

```
import java.io.*;
import java.net.*;

public class EchoClient {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 6000);
            System.out.println("Connected to Echo Server.");
            BufferedReader keyboard = new BufferedReader(
                new InputStreamReader(System.in));
            BufferedReader input = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            PrintWriter output = new PrintWriter(
                socket.getOutputStream(), true);
            while (true) {
                System.out.print("Enter message: ");
                String message = keyboard.readLine();
                output.println(message);
                String response = input.readLine();
                System.out.println("Server Echo: " + response);
                if (message.equalsIgnoreCase("bye")) {
                    break;
                }
            }
            socket.close();
            System.out.println("Connection closed.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:	C:\Users\Bina Computer\OneDrive\Documents\Java>java EchoClient Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8 Connected to Echo Server. Enter message: Hello Server Server Echo: Hello Server Enter message: Java Socket Programming Server Echo: Java Socket Programming Enter message: bye Server Echo: bye Connection closed. C:\Users\Bina Computer\OneDrive\Documents\Java>

Conclusion	The programs successfully demonstrate socket programming in Java. The first exercise establishes a two-way client-server chat using Socket and ServerSocket. The second exercise demonstrates an echo system, where the server receives a message from the client and sends the same message back.